

SAI SUDHEER VISHNUMOLAKALA

813-203-3465 | Atlanta, GA | saisudheerv7@gmail.com | [Linkedin](#) | [Portfolio](#)

SUMMARY

Data Scientist and AI Engineer with **4+ years of experience** building predictive models, GenAI applications, RAG systems, and production analytics workflows across manufacturing, financial services, and research environments. Proven impact includes **\$5.2M in fraud savings**, a **10% reduction in churn**, a **25% lift in sales**, and an **80% reduction in manual processing time**. Skilled at translating complex data and AI solutions into actionable business recommendations and scalable decision workflows. Hands-on expertise in **Python, SQL, machine learning, LLM-powered applications, RAG pipelines, NLP, and model monitoring**.

EXPERIENCE

Data Scientist | ProcessMiner Inc.

June 2025 – Present

Atlanta, GA

- Process Optimization:** Developed ML models for paper manufacturing using high-frequency sensor and lab data to predict COBB quality, optimize chemical usage, reduce process variability, and support real-time production decisions.
- Model Monitoring & Diagnostics:** Built Python/SQL monitoring pipelines across ProcessMiner MySQL schemas to track model accuracy, error reduction, data drift, flat predictions, bias, feature correlations, and actual-vs-predicted performance through automated PDF reports.
- GenAI Decision Support Assistant:** Developed an LLM-powered manufacturing analytics assistant with deterministic SQL routing to answer real-time questions on model status, predictions, machine health, root-cause drivers, and value-capture reports using live ProcessMiner database tables.
- Reinforcement Learning:** Prototyped offline RL control agents using paper mill sensor and lab data, building state-action-reward pipelines with algorithms such as IQL, CQL, and TD3+BC to explore autonomous closed-loop process optimization.

Data Scientist (GRA) | University of South Florida

April 2024 – May 2025

Tampa, FL

- Anomaly Detection:** Built a token-level anomaly detection system with Autoencoders (AUC-ROC: 0.94), using Federated Learning to ensure data privacy across distributed datasets.
- Chatbot:** Developed an intelligent chatbot using web scraping, vector databases, OpenAI embeddings, and LangChain with Retrieval-Augmented Generation (RAG) for accurate, context-aware Q&A.
- ETL Automation:** Migrated Excel workflows to Python-based ETL pipelines with PostgreSQL, reducing processing time by 80% and enabling scalable analytics.
- Developed an **NLP pipeline** to analyze 10,000+ course feedback entries using **BERT embeddings** and sentiment analysis, uncovering actionable themes to improve instructional quality.

Data Scientist | Tiger Analytics

May 2022 – June 2023

Hyderabad, India

- Churn Prediction:** Designed and deployed a predictive churn model (88% precision) that reduced customer attrition by 10%.
- Fraud Detection:** Created real-time fraud detection system using XGBoost, reducing fraud cases by 35% and saving \$5.2M.
- Conducted **A/B testing** to evaluate different sales strategies and promotional displays, leading to a 25% increase in retail sales volume and improved customer acquisition rates.
- Customer Segmentation:** Used unsupervised ML (K-Means, DBSCAN) for segmentation, improving loan conversions by 20%.
- Cross-functional Collaboration:** Liaised with product managers, data engineers, and business stakeholders to translate analytical insights into production-ready solutions and influence key strategic initiatives.

Data Analyst | Tata Consultancy Services

July 2021 – May 2022

Hyderabad, India

- Forecasting:** Developed an LSTM-based time series model to predict store-level sales, cutting forecast error (RMSE) by 7% and improving replenishment accuracy.
- Utilized advanced **SQL** techniques, including **CTEs, window functions**, views, and stored procedures to optimize query performance, enhancing the efficiency of ETL pipelines and supporting complex data operations at scale.
- Re-engineered legacy SPSS Modeler workflows by converting them into optimized SQL scripts, improving performance, maintainability, and enabling integration with modern data pipelines.

EDUCATION

University of South Florida, Muma College of Business

August 2023 – May 2025

Master of Science in Business Analytics and Information Systems

GPA: 3.97

Rajiv Gandhi University of Knowledge Technologies

August 2017 – May 2021

Bachelor of Technology in Electronics and Communication Engineering

GPA: 8.32/10

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, PySpark, TensorFlow, LangChain, Scikit-learn

Machine Learning & AI: Neural Networks, Ensemble Methods, NLP, Time Series Forecasting, Model Interpretability

Statistical Analysis: Hypothesis Testing, Regression, A/B Testing, Causal Inference

Cloud & Big Data: AWS (S3, EC2, SageMaker), Azure (Data Factory, Synapse, ML Studio), Docker, Spark

Generative AI: LLM Applications, RAG Pipelines, Prompt Engineering, Fine-Tuning (PEFT, QLoRA)

Visualization: Power BI, Tableau, Dash, Streamlit, Matplotlib, Seaborn

Other Competencies: PostgreSQL, Excel, Git, MLOps (CI/CD, MLflow), Jira